

(Logical v/s Physical Representation)

Physical

- A hard drive stores data in a series of magnetic fields on a physical disk.
- **Concrete**: concept that focuses on the actual storage of data on physical medium.
- **File System**: concerned with the file system, directory structure, file names and data forms.
- **User-visible**: visible to user and is used to interact with data.
- **Operating system**: managed by operating system, which provides a layer of abstraction between the physical storage & user.
- **Example**: A folder named "Documents" containing file names "Report.Docx" and "Presentation.pptx".

Logical

- The data on the hard drive is organized into files and folders, with names, permissions and other attributes that defines how the data is accessed and used.
- **Abstract** concept that focuses on the organization. Eg structure of data.
- Storage device**: concerned with the storage device, such as a hard drive, solid-state drive or flash drive.
- Bit Level** concerned with Bit level data including the layout of bits on the storage devices.
- Hard drive** ^{ware:} managed by hardware which stores data and retrieve data.
- Example**: A hard drive storing data in a series of magnetic fields on a physical disk.